FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – IV

JEE (Main)-2020

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 75 questions.
- Each subject (PCM) has 25 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each part has only three sections: **Section-A**, **Section-B and Section-C**.

  Section-A (01 – 20, 26 – 45, 51 – 70) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 22, 46 – 47, 71 – 72) *contains 6 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

  Section-C (23 – 25, 48 – 50, 73 – 75) *contains 9 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

Physics

PART – I

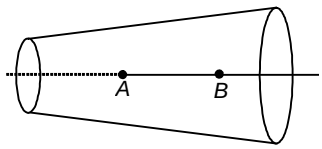
SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

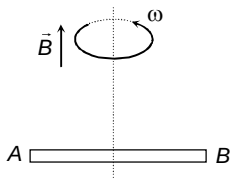
1. If A × B = C, then which of the following statements is wrong-
(A) C ⊥ A (B) C ⊥ B
(C) C ⊥ (A + B) (D) C ⊥ (A × B)

2. A particle moving in a straight line is acted by a force, which works at a constant power and changes its velocity from u to v in passing over a distance x. The time taken will be:
(A) $x\left(\frac{v-u}{v^2+u^2}\right)$ (B) $x\left(\frac{v+u}{v^2+u^2}\right)$
(C) $\frac{3}{2}\,x\left(\frac{v^2-u^2}{v^3-u^3}\right)$ (D) $x\left(\frac{v}{u}\right)$

3. A resistance of frustum shape is shown in figure. If a current *i* passes through the resistance, the electric field at *A* and *B* are related as
(A) $E_A > E_B$
(B) $E_B > E_A$
(C) $E_A = E_B$
(D) there is no relation

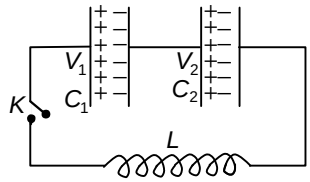


4. A conducting rod of length 2*l* is rotating with constant angular speed ω about its perpendicular bisector. A uniform magnetic field $\vec{B}$ exists parallel to the axis of rotation. The emf induced between two ends of the rod is
(A) $B\omega l^2$ (B) $\frac{1}{2}B\omega l^2$
(C) $\frac{1}{8}B\omega l^2$ (D) zero



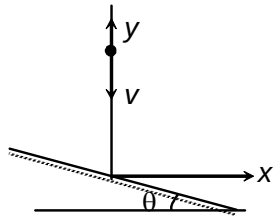
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5. Two capacitors of capacitance C_1 and C_2 are charged to a potential difference of V_1 and V_2 respectively and are connected to an inductor of inductance *L* as shown in the figure. Initially key *k* is open. Now key *k* is closed and current in the circuit starts increasing. When current in the circuit is maximum



(A) charge on both the capacitors is same
(B) induced emf in the inductor is zero
(C) potential difference across both the capacitors is half of the induced emf
(D) electrostatic potential energy stored in both the capacitors is same

6. An object is moving towards a mirror with a velocity *v* as shown in figure. If the collision between the mirror and the object is perfectly elastic, then the velocity of the image after collision with mirror in vector form is



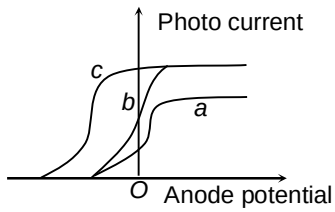
(A) $-v\hat{j}$

(B) $-v\cos 2\theta\hat{j} + v\sin 2\theta\hat{i}$

(C) $-v\hat{i}$

(D) $-v\cos 2\theta\,\hat{j} - v\sin 2\theta\hat{i}$

7. The figure shows the variation of photo current with anode potential for a photo-sensitive surface for three different radiations. Let I_a, I_b and I_c be the intensities and f_a, f_b and f_c be the frequencies for the curves *a*, *b* and *c* respectively;



(A) $f_a = f_b$ and $I_a \neq I_b$

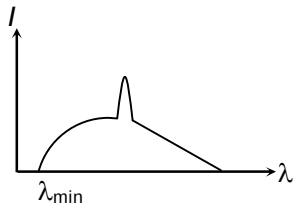
(B) $f_a = f_c$ and $I_a = I_C$

(C) $f_a = f_b$ and $I_a = I_b$

(D) $f_b = f_c$ and $I_b = I_c$

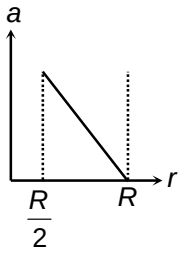
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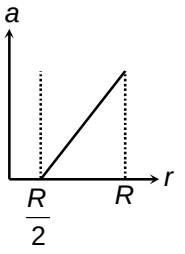
8. An X-ray tube has three main controls.
(i) the target material (its atomic number Z)
(ii) the filament current (I_f) and
(iii) the accelerating voltage (V)
Figure shows a typical intensity distribution against wavelength. Which of the following is incorrect?

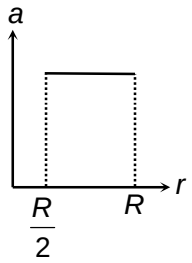


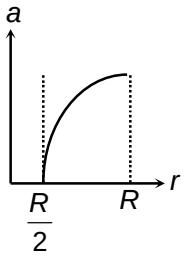
(A) The limit λ_{min} is proportional to V^{-1}
(B) The sharp peak shifts to the right as Z is increased
(C) The penetrating power of X ray increases if V is increased
(D) The intensity everywhere increases if filament current I_f is increased

9. A frictionless tunnel is dug along a chord of the earth at a perpendicular distance $\frac{R}{2}$ from the centre of earth (where R is radius of earth). An object is released from one end of the tunnel. The correct graph, showing the variation of acceleration of particle with its distance r from centre of earth is

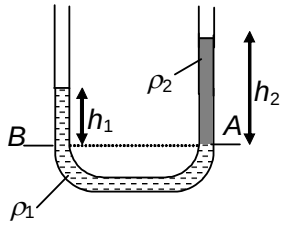
(A)


(B)


(C)


(D)


10. Two immiscible liquids are poured in a U-tube having densities ρ_1 and ρ_2. The ratio of height of the liquids above their interface $\left(h_1 / h_2\right)$ is
(A) directly proportional to their densities
(B) inversely proportional to their densities
(C) directly proportional to square of their densities
(D) equal



Space for Rough work

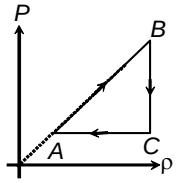
11. Two particles are performing simple harmonic motion with same amplitude and same frequency. When they are at same distance from mean position on opposite sides, their speeds are one fourth of maximum speed. Also at these positions their direction of velocities are opposite, the phase difference between the two simple harmonic motions is

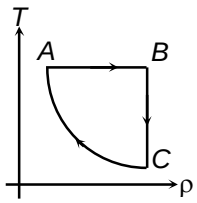


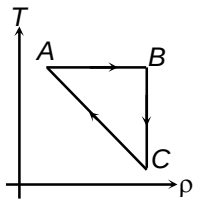
(A) $\frac{\pi}{2}$ (B) π

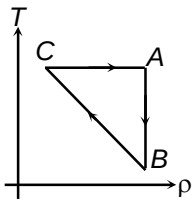
(C) $\frac{3\pi}{2}$ (D) $\frac{5\pi}{4}$

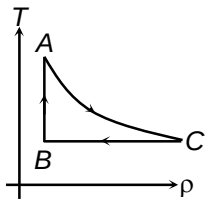
12. Pressure versus density of an ideal gas is shown in the graph. Then the correct temperature versus density of gas graph is



(A)


(B)


(C)


(D)


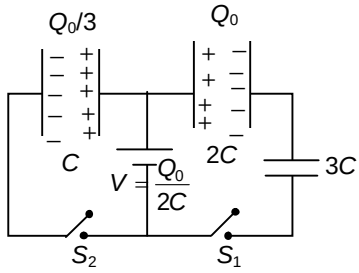
13. P-V diagram of a diatomic gas is straight line parallel to P-axis. The molar heat capacity of the gas in the process will be

(A) 4*R* (B) 2.5 *R*

(C) 3*R* (D) 4*R*/3

Space for Rough work

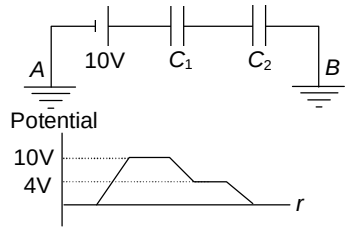
14. In the given circuit, the initial charges on the capacitors are shown in the figure. The charge flown through the switches S_1 and S_2 respectively after closing the switches are



(A) zero, $\frac{Q_0}{6}$ (B) $\frac{Q_0}{5}, \frac{Q_0}{2}$

(C) zero, $\frac{Q_0}{2}$ (D) $\frac{3}{5}Q_0, \frac{Q_0}{6}$

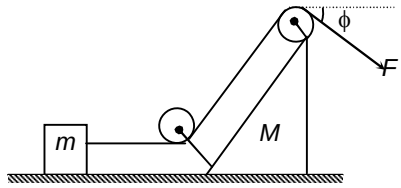
15. Figures shows two capacitors C_1 and C_2 connected with 10 V battery and terminal A and B are earthed. The graph shows the variation of potential as one moves from left to right. Then the ratio of C_1/C_2 is



(A) 5/2 (B) 2/3

(C) 2/5 (D) 4/3

16. In the given arrangement pulleys and string are massless and frictionless and all surfaces are smooth. Find the magnitude of acceleration of wedge of mass M.



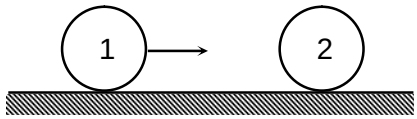
(A) $\frac{F\cos\phi}{M}$

(B) $\frac{F(1-\cos\phi)}{M}$

(C) $\frac{F}{m+M}$

(D) $\frac{F\cos\phi}{m+M}$

17. Ball 1 collides with an another identical ball 2 at rest as shown in figure. For what value of coefficient of restitution e, the velocity of second ball becomes two times that of 1 after collision?

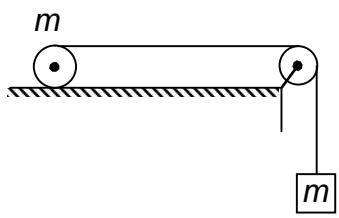


(A) 1/3 (B) 1/2

(C) 1/4 (D) 1/6

Space for Rough work

18. In the given figure a ring of mass *m* is kept on a horizontal surface while a body of equal mass '*m*' attached through a string, which is wounded on the ring. When the system is released the ring rolls without slipping. Consider the following statements and choose the correct option.



(i) acceleration of the centre of mass of ring is $2g/3$

(ii) acceleration of the hanging particle is $4g/3$

(iii) frictional force (on the ring) acts along forward direction

(iv) frictional force (on the ring) acts along backward direction

(A) statement (i) and (ii) only (B) statement (ii) and (iii) only

(C) statement (iii) and (iv) only (D) none of these

19. A uniform rope of length 12 metre & mass 6 kg, is swinging vertically from rigid base. From its free end, one 2 kg mass is attached. At its bottom end one transverse wave is produced of wavelength 0.06 metre. At upper end of rope, wavelength will be-

(A) 1.2 m (B) 0.12 m

(C) 0.12 cm (D) 0.12 mm

20 In a two slit experiment with monochromatic light, fringes are obtained on a screen placed at some distance from the plane of slits. If the screen is moved by 5×10^{-2} m towards the slits, the change in fringe width is 3×10^{-5} cm. If the distance between the slits is 10^{-3} m, calculate the wavelength of the light used.

(A) 7000Å (B) 5000Å

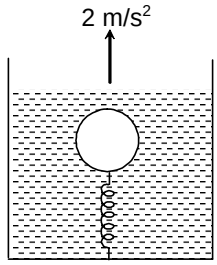
(C) 6000Å (D) 4000Å

Space for Rough work

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

21. A ball of mass 10 kg and density 1 gm/cm^3 is attached to the base of a container having a liquid of density 1.1 gm/cm^3, with the help of a spring as shown in the figure. The container is going up with an acceleration 2 m/s^2. If the spring constant of the spring is 200 N/m, the elongation in the spring is: (in cm)



(A) 2 cm (B) 4 cm
(C) 6 cm (D) 8 cm

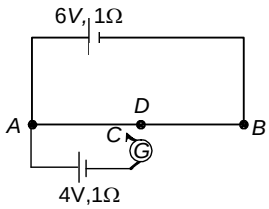
22. A very small sphere of mass 80 g having a charge q is held at a height 9 m vertically above the centre of a fixed non conducting sphere of radius 1 m, carrying an equal charge q. When released it falls until it is repelled just before it comes in contact with the sphere. Calculate the charge q is $7\times K$. The value of K is:
(in μC) [g = 9.8 m/s^2]

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

23. The dimensional formula of a physical quantity x is $\left[M^{-1}L^{3}T^{-2}\right]$. The error in measuring the quantities M, L, and T are 2%, 3% and 4%. The maximum percentage of error that occurs in measuring, the quantity x is:

24. A 6 volt battery of internal resistance 1Ω is connected across a uniform wire AB of length 100 cm. The positive terminal of another battery of emf 4V and internal resistance 1Ω is joined to the point A as shown. Find the distance of point D from A if there is no current through galvanometer. (resistance of *AB* = 5Ω)



25. A brass scale is graduated at 10ºC. What is the true length of a zinc rod which measures 60.00 cm on this scale at 30ºC. Coefficient of linear expansion of brass = 18 × 10^{-6} K^{-1}.

Space for Rough work

Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

26. 0.15 M aqueous solution of potassium sulfate has on osmotic pressure of 8.21 atm at 27°C. Which of the following is correct about the properties of solution?

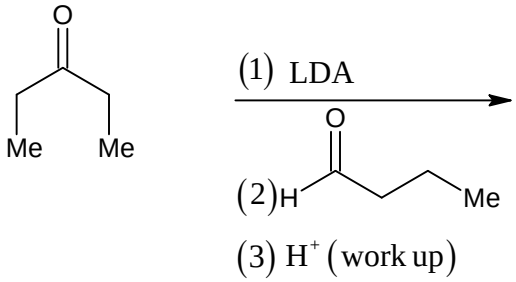
(A) The value of Vant Hoff factor of K_2SO_4 is 3.

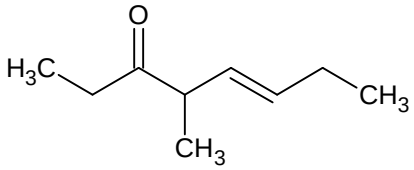
(B) The value of i for K_2SO_4 is less than 3 due to partial dissociation of solute.

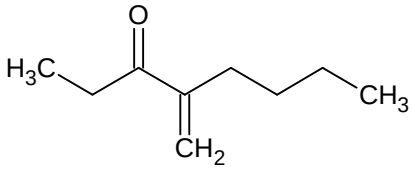
(C) The value of i for K_2SO_4 is more than 3 due to complete dissociation of solute.

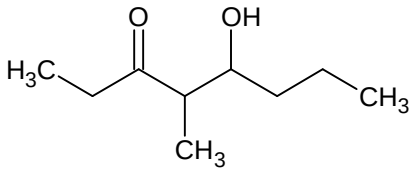
(D) The freezing point o solution would be more than 0°C.

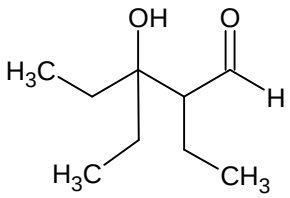
27. Provide the product of the following crossed aldol reaction?



(A) 

(B) 

(C) 

(D) 

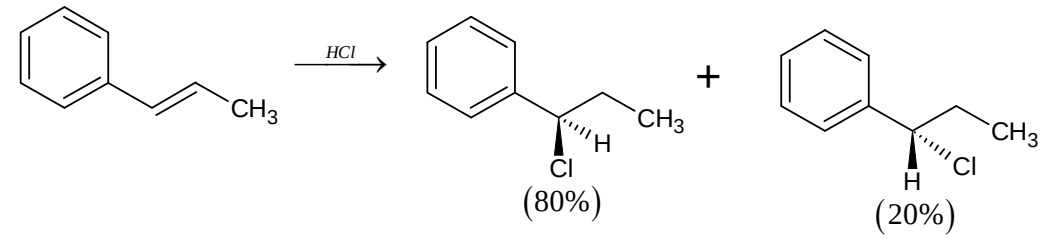
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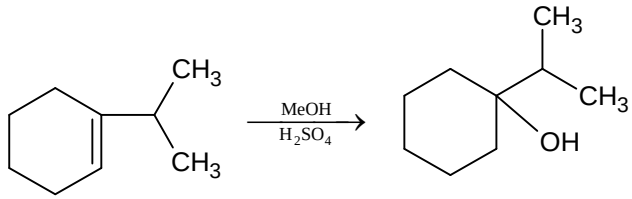
28. A solution is made by equilibrating the two solids barium sulfite $\left(K_{sp}=8\times10^{-7}\right)$ and barium sulfate $\left(K_{sp}=1.0\times10^{-8}\right)$ with water. What are the concentrations of the three ions at equilibrium, if same of each of the solids remain?

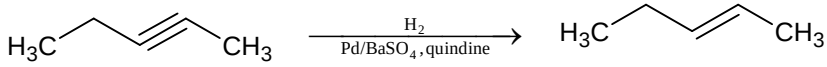
(A) $\left[Ba^{2+}\right]=9\times10^{-4}M$ (B) $\left[SO_4^{2-}\right]=\frac{1}{9}\times10^{-3}M$

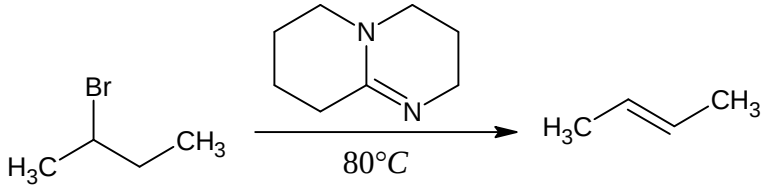
(C) $\left[SO_3^{2-}\right]=\frac{1}{9}\times10^{-3}M$ (D) All are correct.

29. Which of the following reactions are correctly matched with the product?

(A) 

(B) 

(C) 

(D) 

Space for Rough work

30. If we take into account electron spin we effectively double the number of available quantum states for a given set of quantum numbers n, ℓ, m_ℓ. Including spin, how many different quantum states are accessible to an electron with $n = 2$?
(A) 6 (B) 2
(C) 8 (D) 4

31. What is potentially, this single most important property/ability of all matter/material that causes them to react?
(A) Gibbs energy (B) Enthalpy of formation
(C) Entropy of compound (D) Octet rule.

32. What is the change in pressure when 10 grams of molecule A (mw=150 g/mol) reacted with 200ml of 0.20M solution of B and the resulting gas fills a volume of 200ml vessel at 300K?
$2A(s) + B(aq.) \longrightarrow X(g) + 2Y(aq.)$
(A) $5.105 atm$ (B) $4.105 atm$
(C) $2.05 atm$ (D) $2.0547 atm$

33. Which of the following element is not correctly matched with its compound?
(A) The chloride XCl_3 is oxidizing, but XCl is stable and insoluble in water: $T\ell$
(B) The oxide XO_2 has high melting point and is very abundant in nature: Si
(C) X (group 14) forms compounds mainly in +2 oxidation state, through some compounds in the state +4 do exist: Sn
(D) X_4 exists in various allotropic forms and one of the allotrope shows disproportionation reaction with $\mathrm{NaOH} : P$

34. $\mathrm{CoCl_2.6H_2O(s)}$ salt is dissolved in water, which of the following statement is incorrect about this solution:
(A) When 12 M HCl is added to the solution, its color changes to blue.
(B) When the solution is heated, the color of solution becomes blue.
(C) On adding large amount of water, the color of solution becomes pink.
(D) On passing $H_2S(g)$ through the solution, pink colored ppt is formed.

Space for Rough work

35. A volume of 4.0 L of gases A(g) and B(g) react to form solid C(s). During the reaction, the system looses 1150J of heat to the surroundings. The volume of the system is negligibly small at the end of the reaction as the gases disappear under atmospheric pressure. What is the change in internal energy of the system?
(A) $-1150J$ (B) $-744.7J$
(C) $-1555.3J$ (D) $+1150J$

36. $\underset{(1:20)}{Xe+F_2} \xrightarrow[60\text{-}70\,bar]{300^oC} 'A'$ $\quad A+H_2O \longrightarrow B+2HF$

$$A+2H_2O \longrightarrow C+4HF$$

$$A+6H_2O \longrightarrow D+6HF$$

Which of the following statement is correct about the structure of Xe compounds?
(A) Compound B has square pyramidal geometry
(B) Compound C has two lone pairs on control atom.
(C) Compound D is XeO_4.
(D) Compound A has square planer geometry.

37. Which of the ore is correctly matched with its chemical group present?
(A) Cuprite : Sulphate ore (B) Cerussite : carbonate ore
(C) Kainite : Oxide ore (D) Calmine : Chloride ore

38. Which of the following combinations in an aqueous medium will give a red colour or precipitate?
(A) $Fe^{2+} + \left[Fe(CN)_6\right]^{3-}$ (B) $Co^{2+} + SCN^-$
(C) $Cu^{2+} + OH^-$ (D) $Fe^{3+} + SCN^-$

39. The value of CFSE of the complex formed in Brown ring test is:
(A) $-1.8\Delta_o$ (B) $-2.4\Delta_o$
(C) $-0.4\Delta_o$ (D) $-0.8\Delta_o$

Space for Rough work

40. $Mn^{3+} + e^- \longrightarrow Mn^{2+}$ $E^o = 1.50V$

$Fe^{3+} + e^- \longrightarrow Fe^{2+}$ $E^o = 0.77V$

$O_2 + 4H^+ + 4e^- \longrightarrow 2H_2O$ $E^o = 1.23V$

Which of the following statement is correct based on the information given above?

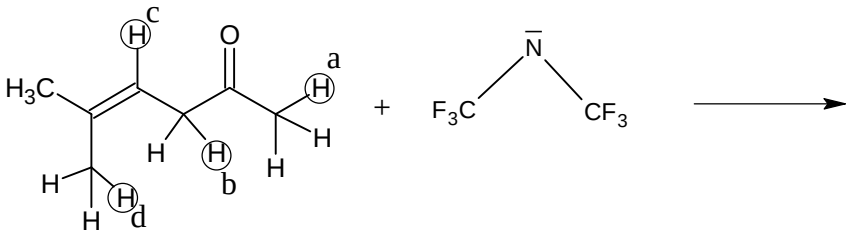
(A) O_2 oxidizes Mn^{2+} to Mn^{3+}.

(B) Mn^{3+} oxidizes H_2O to O_2

(C) O_2 oxidizes both Mn^{2+} to Mn^{3+} and Fe^{2+} to Fe^{3+}

(D) O_2 can not oxidizes Fe^{2+} to Fe^{3+}

41. Which of the following labelled can be removed by base in the above reaction?



(A) a (B) b

(C) c (D) d

42. $A \rightarrow 2B$ mol A consumed/(L.s)=0.1[A]

$B \rightarrow C$ mole C generated /(L.s)=0.2 [B]

The reactor is initially charged with pure A at a concentration of 1.00 mol/L. Which of the following statement is true , about the concentrations at any moment?

(A) [A]+2[B]+[C]=1 mol/L (B) [A]+[B]+[C]=1 mol/L

(C) 2[A]+[B]+[C]=2 mol/L (D) 2[A]+[B]+2[C]=2 mol/L

Space for Rough work

43. Which of the following relationship between radii of Gallium and nitrogen in GaN, and the lattice parameter, a for gallium nitride is correct. Gallium nitride has hcp lattice of nitride ions and Ga^{3+} occupying half of the tetrahedral voids.

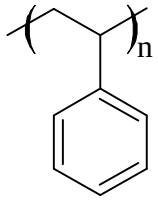
(A) $c=\sqrt{\frac{2}{3}}.r_{N^{3-}}$ (B) $r_{Ga^{3+}} = 0.414\, r_{N^{3-}}$

(C) $a=2\times r_{Ga^{3+}}$ (D) $a=0.89 r_{N^{3-}}$

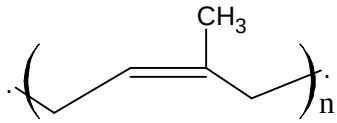
44. Which of the following prospering is incorrect about the complex $\left[Cr\,Br_2\,(en)(NH_3)_2 \right]Cl_2$?

(A) The complex compound can show optical isomerism.
(B) The compound can show ionization isomerism.
(C) The compound can show geometrical isomerism.
(D) The compound forms white ppt (A) with $AgNO_3 sol^n$. These white ppts are soluble in liquid ammonia but insoluble in hypo solution.

45. Which of the following polymer can be easily biodegradable?

(A) 

(B) Bakelite

(C) 

(D) $-(CH_2-CH_2)_n-$

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

46. The number of ions given below for which all octahedral complexes are paramagnetic irrespective of ligand?
$Sc^{3+}, Ti^{3+}, V^{3+}, Cr^{3+}, Mn^{3+}, Fe^{3+}, Co^{3+}, Ni^{3+}$

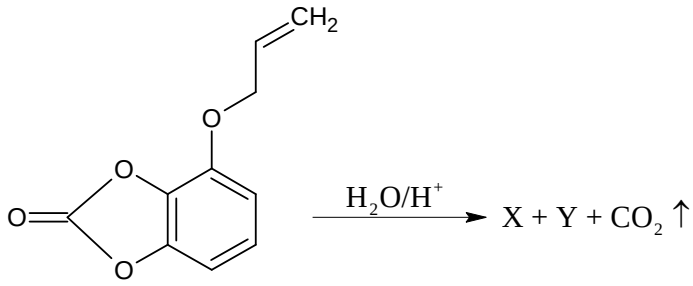
47. H_2O_2 is oxidising agent, how many of ions/molecules can be oxidized by H_2O_2 ?
Fe^{2+} ion, $\left[Fe(CN)_6 \right]^{4-}, NH_2OH, SO_3^{2-}, MnO_4^-, Ce(SO_4)_2, Cl_2, KIO_4$ ...

Space for Rough work

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

48.



$$Y \xrightarrow{\text{Lucas reagent}} \text{turbidity appears immediately}$$

The molecular mass (in g/mol) of compound Y is?
(Consider atomic masses C=12g/mol, H=1 g/mol and O=16g/mol)

49. What is resistance (in a cell with constant 0.2 cm^{-1}) of an 0.18 M solution of acetic acid at 25°C? Given, $K_a = 1.8 \times 10^{-5}$, $\wedge_m^o(CH_3COOH) = 390\ ohm^{-1}\,cm^2\,mol^{-1}$

[Write the answer upto one decimal point in ohm]

50. Amino acid lysine has $p^{Ka_1}(\text{-COOH})=2.20, p^{Ka_2}(\alpha\text{-}NH_3)=8.95$ and $p^{Ka_3}(\text{R-group})=10.53$.

p^I of lysine at 25^oC is?

Space for Rough work

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

51. The order and degree of the differential equations of the family of hyperbolas centred at origin; principal axis along co-ordinate axis and of length $\sqrt{a^2+\lambda}$ and $\sqrt{b^2+\lambda}$ where a, b are fixed real numbers and λ is a real parameter is
(A) 1, 1 (B) 2, 1
(C) 1, 2 (D) 3, 2

52. The nth derivative of the function $f(x) = \frac{1}{1-x^2}$ $(x \in (-1, 1))$ at the point x = 0 where n is even is
(A) 0 (B) n!
(C) $n\ {}^nC_2$ (D) $2\ {}^nC_2$

53. Let $f(x) = (x^n - 1)^{1/n}$ $(x > 0)$, then $\int (f^6(x))^n\,dx$, where $f^n(x)$ denotes $\underset{n \text{ times } f}{fofof.....(x)}$, equals
(A) $\frac{x^{n+1}}{n+1} - 6x + c$ (B) $\frac{x^n}{n} - 64x + c$
(C) $\frac{x^n}{n} - 6x + c$ (D) $\frac{x^{n+1}}{n+1} - 64x + c$

54. If $f(x) = \prod_{k=1}^{2n} \sin(kx)$ where $\prod$ denotes continued product then f′(x) is
(A) $\sum_{k=1}^{2n} (k\cos kx) f(x)$ (B) cosx(2cos2x)(3cos3x)…(2ncos2nx)
(C) $\sum_{k=1}^{2n} (k\cos kx)\sin kx$ (D) $\sum_{k=1}^{2n} (k\cot kx) f(x)$

Space for Rough work

55. The number of points at which the function f(x) = $\begin{cases} \min\left(|x|,\ x^2\right); & -\infty < x < 1 \\ \min\left(2x-1,\ x^2\right); & \text{otherwise} \end{cases}$ is not differentiable, is

(A) 0 (B) 1
(C) 2 (D) more than 2

56. $\lim\limits_{x\to\infty} \dfrac{30+4\sqrt{x}+7\sqrt[3]{x}}{2+\sqrt{4x-7}+\sqrt[3]{6x-2}}$ equals

(A) 15 (B) 2
(C) $\dfrac{3}{2}$ (D) 0

57. $\lim\limits_{n\to\infty}\left(\dfrac{n!}{n^n}\right)^{\sum\limits_{r=1}^{3n}\frac{r}{n^2\sqrt{16n^2+r^2}}}$ equals

(A) e (B) $\dfrac{1}{e^3}$
(C) e^3 (D) $\dfrac{1}{e}$

58. Let f: [k, k + 1, …, 2007] $\to$ [1, 2, 3, …, n] be defined by f(x) = $\left[\dfrac{2007}{x}\right]$ (where [.] denotes the greatest integer function). The maximum value of k such that it is not possible to make f an onto function for any vale of n is

(A) 50 (B) 40
(C) 51 (D) none of these

59. Let f and g be continuous and differentiable functions. If f(0) = f(2) = f(4); f(1) + f(3) = 0; g(0) = g(2) = g(4) = 0 and if f(x) = 0 and g′(x) = 0 do not have a common root, then the minimum number of zeros of f′(x)g′(x) + f(x) g″(x) in [0, 4] is

(A) 3 (B) 4
(C) 5 (D) 2

Space for Rough work

60. Suppose f and g are functions with f″ and g″ existing everywhere. If f(x) g(x) = 2 $\forall$ x and f′(x), g′(x) $\neq$ 0, then $\frac{f''(x)}{f'(x)} - \frac{g''(x)}{g'(x)} = 0$

(A) has at least one real root
(B) may have real roots
(C) has at most one real root
(D) has no real roots

61. Consider a square ABCD of side length 1. Let P be the set of all segments of length 1 with end points on adjacent sides of square ABCD. The mid points of segments in P enclose a region with area A, the value of A is

(A) $\frac{\pi}{4}$
(B) $1 - \frac{\pi}{4}$
(C) $4 - \frac{\pi}{4}$
(D) none of these

62. A curve in the coordinate plane is given by the parametric equations $x = t^2 + t + 2$; $y = t^2 - t + 2$, where $t \geq 0$. The number of straight lines passing through the point (2, 2) which are tangent to the curve is / are

(A) 2
(B) 0
(C) 1
(D) 3

63. $\int_{-5}^{2} \frac{3x^2 + x - 3}{x^2 + 3x + 3} dx$ equals

(A) 15
(B) 21
(C) 30
(D) 42

64. A car is to be driven 200 kms on a highway at an uniform speed of x km/hr (speed rules of the highway require $40 \leq x \leq 70$). The cost of diesel is Rs 30/litre and is consumed at the rate of $100 + \frac{x^2}{60}$ litres per hour. If the wage of the driver is Rs 200 per hour then the most economical speed to drive the car is

(A) 55.5
(B) 70
(C) 40
(D) 80

Space for Rough work

65. Let $0 < \alpha, \beta, \gamma < \frac{\pi}{2}$ are the solutions of the equations cosx = x, cos(sinx) = x and sin(cosx) = x respectively then
(A) $\alpha < \beta < \gamma$ (B) $\gamma < \alpha < \beta$
(C) $\beta < \alpha < \gamma$ (D) $\gamma < \beta < \alpha$

66. If f(x) is a real valued function satisfying $f(x - y) = \frac{f(x)}{f(y)}$ $\forall$ x, y $\in$ R and if f′(0) = p; f′(5) = q, then f′(–5) equals
(A) $\frac{p^2}{q}$ (B) $\frac{p}{q}$
(C) $\frac{q}{p}$ (D) q

67. $\bar{a}, \bar{b}, \bar{c}$ are the edges of cube of unit length and $\hat{r}$ be a unit vector in the cube, then $|\hat{r} \times \bar{a}|^2 + |\hat{r} \times \bar{b}|^2 + |\hat{r} \times \bar{c}|^2$ is
(A) 0 (B) 1
(C) 2 (D) 3

68. The number of ways in which 2n white and 2n black balls can be arranged such that no consecutive 'n' white balls are together are
(A) ${}^{2n+1}C_2 + {}^{4n}C_{2n}$ (B) ${}^{2n+1}C_2 - {}^{2n+1}C_1 \, {}^{3n}C_n \, (-1)^n + {}^{4n}C_{2n}$
(C) ${}^{2n+1}C_2 + (-1)^n \, {}^{2n+1}C_1 \, {}^{3n}C_n + {}^{4n}C_{2n}$ (D) ${}^{2n+1}C_2 + (-1)^n \, {}^{3n}C_n \, {}^{2n+1}C_1$

69. If the equation $x^2 + 2|a|x + 4 = 0$ has integral roots, then minimum value of a is
(A) 4 (B) –5/2
(C) 0 (D) –4

70. If line through P(a, 2) meets the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ at A and D and meets the axes at B and C, so that PA, PB, PC, PD are in G.P., then
(A) a $\in$ [6, ∞) (B) a $\in$ R
(C) a $\in$ $(-\infty, -6] \cup [6, \infty)$ (D) none of these

Space for Rough work

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

71. If matrix $A = \begin{bmatrix} 3 & 2 & 1 \\ 6x^2 & 2x^3 & x^4 \\ 1 & a & a^2 \end{bmatrix}$, then $\frac{d^2}{dx^2}(|A(adjA)|^{1/3})$ at x = a is _____.

72. The system of equation $2x+3y-z=0$, $3x+2y+kz=0$, $4x+y+z=0$ have a set of nonzero integral solution then the least positive value of z is _____

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

73. The maximum integral value of the function $f(x)=\int_{\pi/2}^{x}(\cos t-2\sin t)dt$ in the interval $\left[\frac{\pi}{2},\pi\right]$ is _____

74. The differential equation $dy = x\tan y(\sin y-1)dx$ has solution $y=\sin^{-1}\left(\frac{1}{k+ce^{x^2/2}}\right)$ then $k =$ _____

75. Y-axis is a tangent to one ellipse with foci (2, 0) and (6, 4). If the length of the major axis is x then $\frac{[x]}{5}$ =

Space for Rough work